

IT Helpdesk – 18 Day Windows Server & Active Directory Home Lab

A fully hands-on IT lab that simulates a real-world helpdesk enterprise that uses VirtualBox and a Windows 2022 Server.

Project Overview

IT environment includes hands-on experience with Active Directory, Networking, Security Policies, Scripting (Windows PowerShell ISE), fully functioning ticketing system (OSTicket), and log analysis. This lab directory correlates with the daily responsibilities of IT Helpdesk or similarly related roles.

Domain: alexlang-helpdesk.com

OS: Windows Server 2022+ Windows 11 Pro

Hypervisor: VirtualBox

Skills & Technologies

Category	Tools / Concepts
Virtualization	VirtualBox, VM provisioning, VHD management, Guest Additions, network adapter modes
Windows Server	Windows Server 2022, Server Manager, SConfig, RDP
Active Directory	AD DS, domain promotion, OU structure, user & group management
DNS	Forward/Reverse Lookup Zones, A, CNAME, SOA, NS, SRV records
DHCP	Scope configuration, lease management, IP exclusions
Group Policy	GPO creation, password/lockout policy, software deployment, printer policy

PowerShell	Bulk user creation, password reset scripts, system info automation
NTFS Permissions	Inheritance, security groups, share permissions, effective access auditing
Ticketing System	osTicket, XAMPP/Apache, MySQL/phpMyAdmin
Patch Management	Action1 endpoint agent, vulnerability scanning, update deployment
Backup & Recovery	Shadow Copies, previous version restoration, Windows Server Backup
Event Viewer	Log filtering, custom views, Event IDs 4625/4740, .evtx export



Day-by-Day Summary

Day 1 — VM Setup: Created a Windows Server 2022 using VirtualBox, resolved a display crash (switched to VMSVGA), and configured initial VM/Server settings via SConfig.

Day 2 — Active Directory: Installed AD(Active Directory)/DS(Domain Services), promoted server to a Domain Controller, created forest alexlang-helpdesk.com, and used some command line Networking commands (whoami, net user, net accounts).

Day 3 — Guest Additions: Installed VirtualBox Guest Additions and configured a host-to-VM shared folder.

Day 4 — Users & Groups: Created a few domain users (John Stevens, Julie Foxburrow, Sully Johnson), a security group named IT_Help, and explored user property tabs and privilege settings.

Day 5 — Workstation Deployment: Installed two Windows 11 Pro VMs simultaneously, used OOBE\BYPASSNRO to bypass Microsoft account requirements, and resolved a second VM boot failure via regedit.

Day 6 — Domain Join & Networking: Configured static IPs, pointed DNS to the DC, joined both workstations to the domain, used `ping` and `ipconfig /all` to verify, and troubleshooted a local vs. domain login issue.

Day 7 — RDP & Group Policy: Enabled RDP, added domain user to RDP users, connected remotely to test it, configured password/lockout policies via GPO, (Group Policy Object) and verified with `gpresult /R` and `net user` commands.

Day 8 — Patch Management: Deployed an Action1 agent to the DC (Domain Controller), identified 3 critical vulnerabilities and 7 missing updates on first scan, and ran important updates to remediate these vulnerability alerts.

Day 9 — Shared Folders: Created Tech and HelpDesk shares via the Server Manager, and configured share-level permissions restricting default access.

Day 10 — Advanced GPO: Explored full GPME (Group Policy Management Editor) structure, configured screen saver timeout/display policies, and printer policies (browse URL, prevent deletion).

Day 11 — DHCP: Installed DHCP Server role, created a custom scope with IP range/lease/gateway, switched workstations to DHCP, and disabled the hypervisor's built-in DHCP to avoid IP conflicts.

Day 12 — DNS: Reviewed AD auto-generated records (SOA, NS, SRV), created a custom A record, CNAME alias (`files.alexlang-helpdesk.com`), and a Reverse Lookup Zone.

Day 13 — PowerShell Automation: Wrote scripts for bulk user creation (`createUser.ps1`), forced password reset (`changeUser.ps1`), and system info retrieval (`SystemInfo.ps1`), verified scripts results in AD.

Day 14 — osTicket: Installed a XAMPP Server and osTicket, configured phpMyAdmin database and admin user, created departments (IT Support,

Network Ops) and help topics, and troubleshoot config file permissions and email validation errors.

Day 15 — NTFS Permissions: Built departmental folder structure (HR, IT, Sales, Public). Created AD security groups per department, assigned NTFS permissions, and audited access using the Effective Access tab.

Day 16 — Software Deployment: Deployed 7-Zip via GPO software installation policy using a domain-accessible MSI share and ran gpupdate /force — application auto-installed on reboot.

Day 17 — Backup & Recovery: Set up Shadow Copies on the C: drive (daily at 2PM), deleted a test file and successfully restored it via latest version of C drive, and troubleshoot a full backup failure due to insufficient VM disk space.

Day 18 — Event Viewer & Log Analysis: Explored Windows Security logs. Triggered Event ID 4625 (failed logon) intentionally. Created a custom view for IDs 4625/4740. Exported logs as .evtx to simulate escalation workflow.

Crucial Takeaways

Real World Troubleshooting – Most of the days included some kind of issues (Software Bugs, Internet Issues, Security Migrations, etc) that each problem had to be troubleshoot through and didn't have a linear solution and required multiple troubleshooting steps

Prioritizing Security (following Zero Trust Architecture) - Maximizing security whenever possible including: enforcing password complexity, lockout thresholds, NTFS permissions, following principle of least privilege, and audit log monitoring

Fully hands on Environment Scope – Setup a fully networked domain with users, policies, shared resources, a full-fledged ticketing system, and monitoring

PowerShell Scripting – Powershell scripts were used for automating repetitive tasks such as user creation, and password resets

Network Infrastructure Administration – Implemented and managed a full network including configuring DHCP scopes and managing DNS records

Certifications in progress during project:

Completed CompTIA A+

In progress CompTIA Network+